

Predicates & Quantifiers

$$x > 3$$

↪ cannot be expressed in propositional logic

predicate logic is propositional logic with variables.

" x is greater than 5"

subject predicate

$P(x)$ → propositional function

↙ notes the predicate ↘ subject

Example Let $P(x)$ denote " $x > 3$ ".

What are truth values of $P(7)$, $P(1)$, $P(-8)$?

T F
F

Example Let $P(x, y)$ denote " $x + y \geq 6$ ".

$$P(3, 3) \quad T$$

$$P(7, -5) \quad F$$

$$P(5, 9) \quad T$$

Quantifiers

→ one way of assigning values to variables in a propositional function.

predicate calculus → area of logic dealing with predicates and quantifiers.

The universal quantifier

' $\forall$ ' means 'for all'

' $\forall x$ ' means 'for all values of x in the domain under consideration'

Example

Let $P(x)$ be " $x+1 > x$ ".

What is the truth value of $\forall x P(x)$, where the domain under consideration is the set of all real numbers.

Ans: TRUE

Example: $P(n)$ is " $x < 2$ "

domain: set of real numbers.

$\forall n, P(n)$ $\rightarrow$ what is the truth value?

Ans: FALSE

Existential Quantifier

' $\exists$ ' means 'there exists'

$\exists x$ means there exists an x in the domain under consideration.

Examples

$P(n)$ is " $x < 2$ "

What is the truth value of $\exists x, P(n)$?

Domain: set of natural numbers

Ans: TRUE

Example: $P(n) \Rightarrow$ " $x^2 > 10$ ", Domain: $\{1, 2, 3, 4\}$

What is the truth value of " $\exists x P(n)$ "

What is the truth value of " $\exists x, P(x)$ "

Ans: TRUE.

Precedence of Quantifiers

$\forall$ and $\exists$ have higher precedence than all logical operators in propositional calculus.

Logical Equivalence involving quantifiers

$$(1) \forall x (P(x) \wedge Q(x)) \equiv \forall x (P(x)) \wedge \forall x (Q(x))$$

$$(2) \exists x (P(x) \vee Q(x)) \equiv (\exists x (P(x))) \vee (\exists x (Q(x)))$$

Negating quantified expressions

$$(1) \neg (\forall x (P(x))) \equiv \exists x (\neg P(x))$$

Suppose domain = $\{a_1, a_2, \dots, a_k\}$

$$\equiv \neg (P(a_1) \wedge P(a_2) \wedge \dots \wedge P(a_k))$$

$$\begin{aligned}
 & (\neg P(a_1) \wedge \neg P(a_2) \wedge \dots \wedge \neg P(a_k)) \\
 & \equiv (\neg P(a_1)) \vee (\neg P(a_2)) \vee \dots \vee (\neg P(a_k)) \quad \text{(De Morgan)} \\
 & \equiv \exists x \neg P(x)
 \end{aligned}$$

$$(2) \quad \neg (\exists x P(x)) \equiv \forall x (\neg P(x))$$

Example (What are the negations of the following statements?)

a. There is an honest politician.

Ans: No politician is honest.

b. $\forall x, (x^2 > x)$

Ans: $\exists x, (x^2 \leq x)$

Translating from english into logical expressions

Q. "Every student in this class has studied calculus."

Ans: Let $C(x)$ denote "x has studied"

Ans: Let $C(x)$ denote "x has studied calculus"

$\forall x, C(x)$, where domain is the set of students in this class

What if domain was set of all students in the institute?

$S(x)$ denotes "x is a student in this class"

$$\forall x, S(x) \rightarrow C(x)$$

$$\forall x, S(x) \wedge C(x) \quad \times$$

Q. "All lions are fierce" $\rightarrow \forall x, L(x) \rightarrow F(x)$

$$\times \exists x, (L(x) \vee C(x))$$

"Some lions do not drink coffee"

$$\times \exists x, L(x) \rightarrow C(x) \quad \hookrightarrow \exists x, (L(x) \wedge C(x))$$

"Some fierce creatures do not drink coffee"

$$\hookrightarrow \exists x, (F(x) \wedge C(x))$$

Let domain be the set of all creatures.

$L(x)$: "x is a lion"

$C(x)$: "x does not drink coffee"

$F(x)$: "x is a fierce creature"

Nested quantifiers

Order matters

$\forall x \exists y P(x, y)$ is not necessarily the same
as $\exists y \forall x P(x, y)$

Example:

Domain: real numbers

$\forall x, \exists y, x + y = 0$ — TRUE

$\exists y, \forall x, x + y = 0$ — FALSE

Translating English to logical expressions involving nested quantifiers

Q. "The sum of two positive integers is always positive"

Ans: $\forall x, \forall y, ((x > 0) \wedge (y > 0)) \rightarrow (x + y > 0)$
Domain: integers

Q. Every real number except zero has a multiplicative inverse

Ans: $\forall x, (x \neq 0) \rightarrow (\exists y, (x \cdot y = 1))$
 $\forall x \exists y (x + x) \rightarrow \dots$

$$\forall x, (x \neq 0) \rightarrow (\exists y, (x \cdot y = 1))$$

$$\forall x, \exists y (x \neq 0) \rightarrow (x \cdot y = 1)$$

Domain: real numbers

Reverse translations

Example 11

$F(x) :-$ "x is a female"

$P(x) :-$ "x is a parent"

$M(x, y) :-$ "x is the mother of y"

$$\forall x (\exists y (F(x) \wedge P(x)) \rightarrow M(x, y))$$

Domain: all people

Example 12 Everyone has exactly one best friend

$B(x, y) :-$ "y is the best friend of x"

$$\forall x, \exists y (\exists z (B(x, y) \wedge ((z \neq y) \rightarrow \neg B(x, z))))$$

such that

Domain: all people

Example 14 $\forall x \exists y (xy = 1) \rightarrow$ Negate this

$$\exists x \forall y (xy \neq 1)$$

Ex. 1. is "There does not a number x such that

Example 15. "There does not a woman who has taken a flight on every airline in the world"

Express this in predicate logic.

Domain for x : all women

Domain for y : all airlines

$$\forall x \exists y \neg F(x, y)$$

x has taken a flight on airline y

Introduction to proofs

Direct proof

To show: $\forall x P(x) \rightarrow Q(x)$

We start by taking an arbitrary t from the domain of x and assume $P(t)$ is true. Using a series of deductions (or various rules of inference), we finally infer that $Q(t)$ is true.

Example! Give a direct proof to the theorem

"if n is an odd integer, then n^2 is odd"

"if n is an odd integer, then n^2 is odd"

Proof: $P(n)$: n is odd

$Q(n)$: n^2 is odd

To show: $\forall n \quad P(n) \rightarrow Q(n)$

Domain: integers

Take an arbitrary integer n , Assume $P(n)$.

Then, $n = 2k+1$, for some integer k .

Then, $n^2 = (2k+1)^2$

$$= 4k^2 + 4k + 1$$

$$= 2(2k^2 + 2k) + 1$$

$$= 2t + 1, \quad \text{where} \\ t = 2k^2 + 2k \\ \text{is an integer.}$$

Thus, n^2 is odd.

Hence proved.

Example 2: Give a direct proof to

"If m and n are perfect squares,

then mn is also a perfect square"

Soln: $\forall n \forall m \quad P(n) \wedge P(m) \rightarrow P(nm)$

Soln.

$$\forall x \forall y \quad p(x) \wedge p(y) \rightarrow p(xy)$$

$$m = a^2, \text{ for some integer } a$$

$$n = b^2, \text{ " " " " } b.$$

$$mn = a^2 b^2 = (\underbrace{ab}_{\text{integer}})^2$$

□

Proof by Contraposition

To show $p \rightarrow q$, it is enough to show
 $\neg q \rightarrow \neg p$.

Example: Let n be an integer.
Prove that if ~~n is an integer~~
~~and~~ $3n+2$ is odd, then n is odd

Proof: Let n be an integer. "If n is even,
then $3n+2$ is even"

$$\text{Let } n = 2k, \text{ for some integer } k.$$

$$\begin{aligned} \text{Then, } 3n+2 &= 3(2k) + 2 \\ &= 6k+2 \end{aligned}$$

$$\begin{aligned}
 &= 6k+2 \\
 &= 2(3k+1)
 \end{aligned}$$

→ Then, it is even,

□

Example If $n = ab$, where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Proof: Let a and b be true integers.

"If $a > \sqrt{n}$ and $b > \sqrt{n}$, then $n \neq ab$ "

Vacuous proofs or trivial proofs

Statements like $\forall n, P(n) \rightarrow Q(n)$

The case where $P(n)$ is FALSE, the above statement is vacuously true.

Example Prove that the sum of two rational numbers is a rational number.

Proof: Let $r_1 = \frac{a}{b}$, $r_2 = \frac{c}{d}$,

Proof.

$$\text{Let } r_1 = \frac{a}{b}, \quad r_2 = \frac{c}{d},$$

where a, b, c, d are integers with $b \neq 0, d \neq 0$,

$$r_1 + r_2 = \frac{a}{b} + \frac{c}{d}$$

$$= \frac{ad + bc}{bd}$$

integer

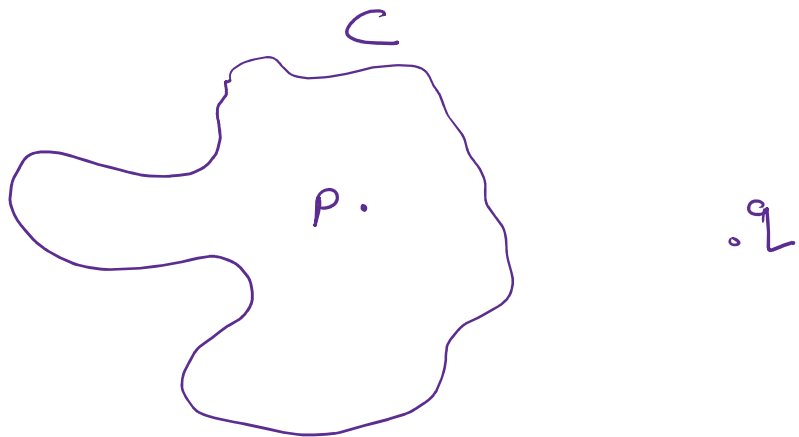
integer

modulo arithmetic

$$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}.$$

$$2 \times 3 = 0$$

Jordan curve theorem



Fermat's Last Thm

$$x^r + y^r = z^r$$

does not have an integer soln

does not have an integer soln
for x, y, z with $x > 0, y > 0, z > 0$ when $r > 2$.

Andrew Wiles $x=3, y=4, z=5, r=2$
 $3^2 + 4^2 = 5^2$

↳ documentary on Fermat's Last Theorem by
Simon Singh.

Q. Let n be an integer. Prove that if n^2 is
odd, then n is odd.

↳ show if n is even, then n^2 is even

Proof by Contradiction

To show: P is TRUE.

Assume P is FALSE. That is, $\neg P$ is TRUE.

Using a series of logical deductions, show that Q
is TRUE. which we know is a contradiction
(a stmt which is always FALSE).

$$\neg P \rightarrow Q$$

$$\frac{72}{\frac{0 \rightarrow p}{\underline{\underline{\quad}}}}$$

Example Show that at least 4 of any 22 days must fall on the same day of the week.

Soln. Assume the stmt is FALSE.

Mondays : ≤ 3

Tuesdays : ≤ 3

;

Sundays : ≤ 3

$3 \times 7 = 21$
days given ≤ 21
which is a contradiction.

Example. Prove that $\sqrt{2}$ is irrational.

Proof: Proof by contradiction.

Assume, for the sake of contradiction, that

Assume, for the sake of contradiction, that $\sqrt{2}$ is rational.

Let $\sqrt{2} = \frac{a}{b}$ (assume a & b have no common factors)

Then,

$$2 = \frac{a^2}{b^2} \quad \text{--- (1)}$$

$$\text{Or } a^2 = 2b^2$$

Thus, a^2 is even. This means, a is even.

Or, $2 \mid a$. divides. (2)

Thus, $a = 2k$, for some integer k .

Thus, eqn (1) becomes,

$$2 = \frac{a^2}{b^2} = \frac{(2k)^2}{b^2} = \frac{4k^2}{b^2}$$

$$\text{Or } b^2 = 2k^2.$$

Thus b^2 is even. Therefore, b is even.

Or, $2 \mid b$. (3)

(3)

From (2) and (3), we conclude that
2 divides a and 2 divides b .

This contradicts the fact that a and
 b are coprimes.

□

Example. Give a proof by contradiction
to

"if $3n+2$ is odd, then n is odd"

Proof: Assume, for the sake of contradiction,
that $3n+2$ is odd and n is even.

Let $n = 2k$, for some $k \in \mathbb{Z}$.

$$\begin{aligned}\text{Then, } 3n+2 &= 3(2k) + 2 \\ &= 6k + 2 \\ &= 2(3k+1)\end{aligned}$$

This contradicts the fact that $3n+2$ is
odd.

Proof of Equivalence

Example. Prove $\iff$ "or" if and only if "if"

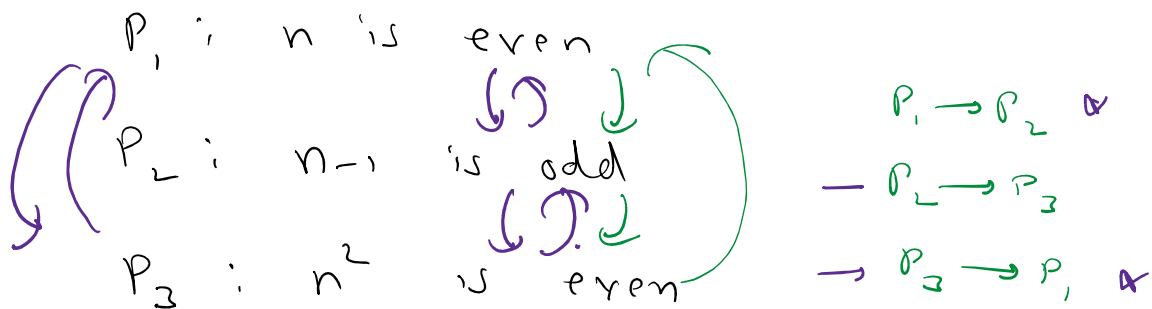
"if n is a +ve integer, then n is odd if and only if n^2 is odd"

Proof. Let n be a +ve integer.

$\rightarrow$ (a) if n is odd, then n^2 is odd.

$\Leftarrow$ (b) if n^2 is odd, then n is odd.

Example. Show that the following statements about integer n are equivalent.



Proof.

Counterexamples

Example: Show that the following stat is FALSE.

"Every true integer is the sum of the squares of two integers"

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Exhaustive proofs or proofs by cases

To show.

$$(P_1 \vee P_2 \vee P_3 \vee \dots \vee P_k) \rightarrow Q$$

Instead we show,

$$(P_1 \rightarrow Q) \wedge (P_2 \rightarrow Q) \wedge \dots \wedge (P_k \rightarrow Q)$$

Example: Prove that $(n+1)^3 \geq 3^n$, when $1 \leq n \leq 4$.

Example: Prove that the only ...

Example. Prove that the only consecutive
the integers not exceeding 100 that
are perfect powers are 8 and 9.

$$\begin{matrix} 11 \\ 2^3 \end{matrix} \quad \begin{matrix} 11 \\ 3^2 \end{matrix}$$

Constructive existence proofs

Example. Show that there is a true
integer that can be written as sum
of cubes of the integers in two different
ways.

$$1729 = 10^3 + 9^3 = 12^3 + 1^3$$

Non-constructive existence proofs

Example. Show that there exist
irrational numbers x and y s.t.
 x^y is rational

Proof.

1. ...

Partition no.

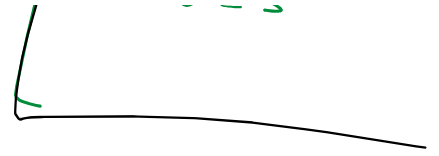
$$\begin{matrix} 3 = 3+0 \\ \quad 2+1 \\ \quad 1+1+1 \end{matrix} \quad \left. \vphantom{\begin{matrix} 3 = 3+0 \\ 2+1 \\ 1+1+1 \end{matrix}} \right\} 3$$

$$\begin{matrix} 4 = 4 \\ = 3+1 \\ = 2+2 \\ = 2+1+1 \\ = 1+1+1+1 \end{matrix} \quad \left. \vphantom{\begin{matrix} 4 = 4 \\ = 3+1 \\ = 2+2 \\ = 2+1+1 \\ = 1+1+1+1 \end{matrix}} \right\} 5$$

$p(n)$ → no. of
ways you
can write n .
 $p(3) = 3$
 $p(4) = 5$

very

Take $x = \sqrt{2}$, $y = \sqrt{2}$



Case 1 $\sqrt{2}^{\sqrt{2}}$ is rational. Then, we are done.

Case 2. Suppose $\sqrt{2}^{\sqrt{2}}$ is irrational.

Then, $x = \sqrt{2}^{\sqrt{2}}$, $y = \sqrt{2}$

$$x^y = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^2 = 2 //$$

Paul Erdős